

Hanzhen Wang

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PERSONAL INFORMATION

Name: Hanzhen Wang
Date of Birth: Jan. 2005
Year: Senior (4rd Year Undergraduate)
Institution: Shanghai Jiao Tong University (SJTU)

EDUCATION

2023.09 - Now **Shanghai Jiao Tong University (SJTU)** **Shanghai, China**
Undergraduate student in School of Automation and Intelligent Sensing
Major in Automation

- Selected Core Courses: Introduction to Control Systems: 98/100, Linear and Nonlinear Programming: 97/100, Robotics: 97/100, Machine Learning and Knowledge Discovery: 96/100.

RESEARCH

My research focuses on **Efficient Artificial Intelligence Models**, optimizing from both algorithmic and system perspectives for both cloud and edge scenarios. The detailed topics are as follows.

- Large Language Model Optimization**
Sparse attention and KV Cache management; Speculative decoding algorithm design; Training of ultra-large models with speculative decoding.
- Embodied AI**
Visual token compression and action-aware scheduling; Agent-driven Embodied Inference and RL training.

PUBLICATIONS (*EQUAL CONTRIBUTION, †CORRESPONDING AUTHOR)

- [C1] **Hanzhen Wang***, Jiaming Xu*, Jiayi Pan, Yongkang Zhou, Guohao Dai†, “SpecPrune-VLA: Accelerating Vision-Language-Action Models via Action-Aware Self-Speculative Pruning”, to appear in *International Conference on Machine Learning (ICML)*, 2026.
- [C2] Jiaming Xu, Jiayi Pan, **Hanzhen Wang**, Yongkang Zhou, Jiancai Ye, Yu Wang, Guohao Dai†, “SpeContext: Enabling Efficient Long-context Reasoning with Speculative Context Sparsity in LLMs”, to appear in *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2026.
- [C3] Yongkang Zhou, Jiaming Xu, Jiayi Pan, **Hanzhen Wang**, Tianlang Zhao, Guohao Dai†, “MTPMoE: Exploiting Activation Skewness in Single-GPU MoE Multi-Token Prediction via Parallel Dense-Sparse Computing”, to appear in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2027.

[C4] Jiaming Xu, **Hanzhen Wang**, Tianlang Zhao, Yongkang Zhou, Yifan Jiao, Chengze Yuan, Guohao Dai[†], “FlashMoD: Approximate Dynamic Programming Helps Efficient Mixture of Depth in Large Language Models on Edge Devices”, (Under review), 2026.

[C5] Jiaming Xu, **Hanzhen Wang***, Tianlang Zhao, Jiewen Xiao, Chengze Yuan, Jiayi Pan, Yongkang Zhou, Jiancai Ye, Guohao Dai[†], “DLite: History-Free Speculative Decoding for Agentic AI”, (Under review), 2026.

AWARDS AND HONORS

2025	Undergraduate Merit Scholarship , Shanghai Jiao Tong University
2025	Third Prize , Shanghai Regional Division, National Undergraduate Electronic Design Contest.
2024, 2025	Outstanding Student Award , Shanghai Jiao Tong University. (Top 3% of Undergraduates).
2024	National First Prize , China Robot Competition and RoboCup China Open.

PROJECT EXPERIENCE

Airport Ground Support Equipment (GSE) Tracking System

Under Prof. Zhenyu Huang, Shanghai Jiao Tong University *Collaboration with China Eastern Airlines, 2025*

- Project Summary: An IoT(Internet of Things) -enabled tracking system for unpowered airport ground vehicles (e.g., baggage carts). It helps to determine real-time location and usage duration. I am responsible for the cloud computing.
- Main contribution:
 1. **At the Network Layer:** Designed and implemented the backend service using **Java Spring Boot**, enabling data ingestion from edge devices and integration with cloud infrastructure. Built high-throughput, low-latency network communication layer using **Netty**, supporting concurrent connections from over 500 vehicles.
 2. **At the algorithm level:** Proposed a dynamic filtering algorithm that leverages temporal correlation of RSSI (Received Signal Strength Indicator) sequences to denoise raw signals and improve location accuracy under multipath interference.
 3. **Cloud Database:** Deployed time-series data pipeline to MySQL-based cloud database, enabling real-time monitoring of vehicle positions and usage duration.

INTERNSHIP EXPERIENCE

Research Intern at **Dots Infra, RedNote (xiaohongshu)** Sept. 2026 – Present
Supervisor: Dr. Zheng Qu
Research Direction: Speculative Decoding Design and Training in LLM

TECHNICAL SKILLS

Programming Languages: Proficient in C++, Java and Python.
Frameworks & Tools: Java Spring Boot and Netty, llama.cpp (LLM deployment in AIPC).